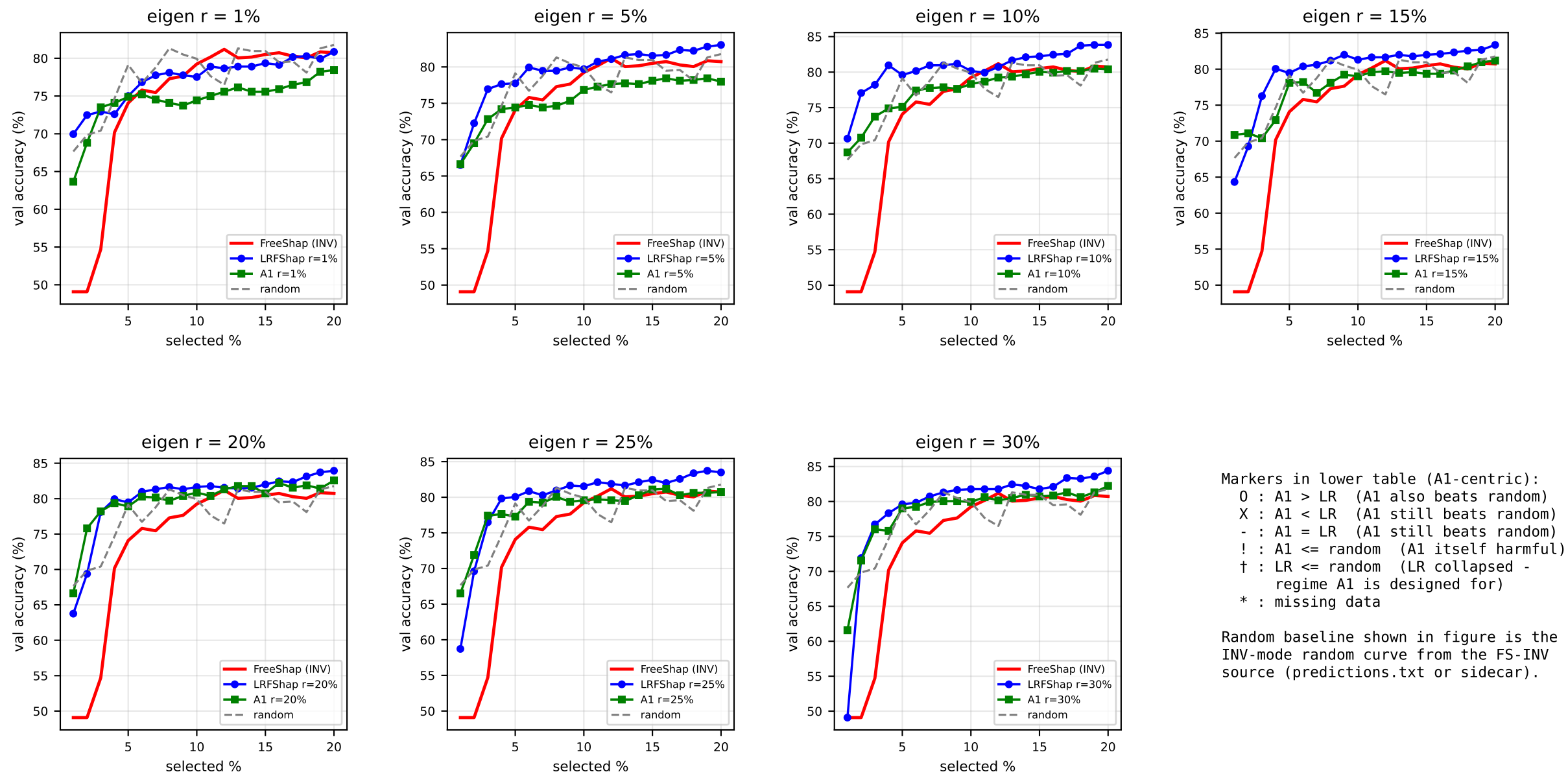
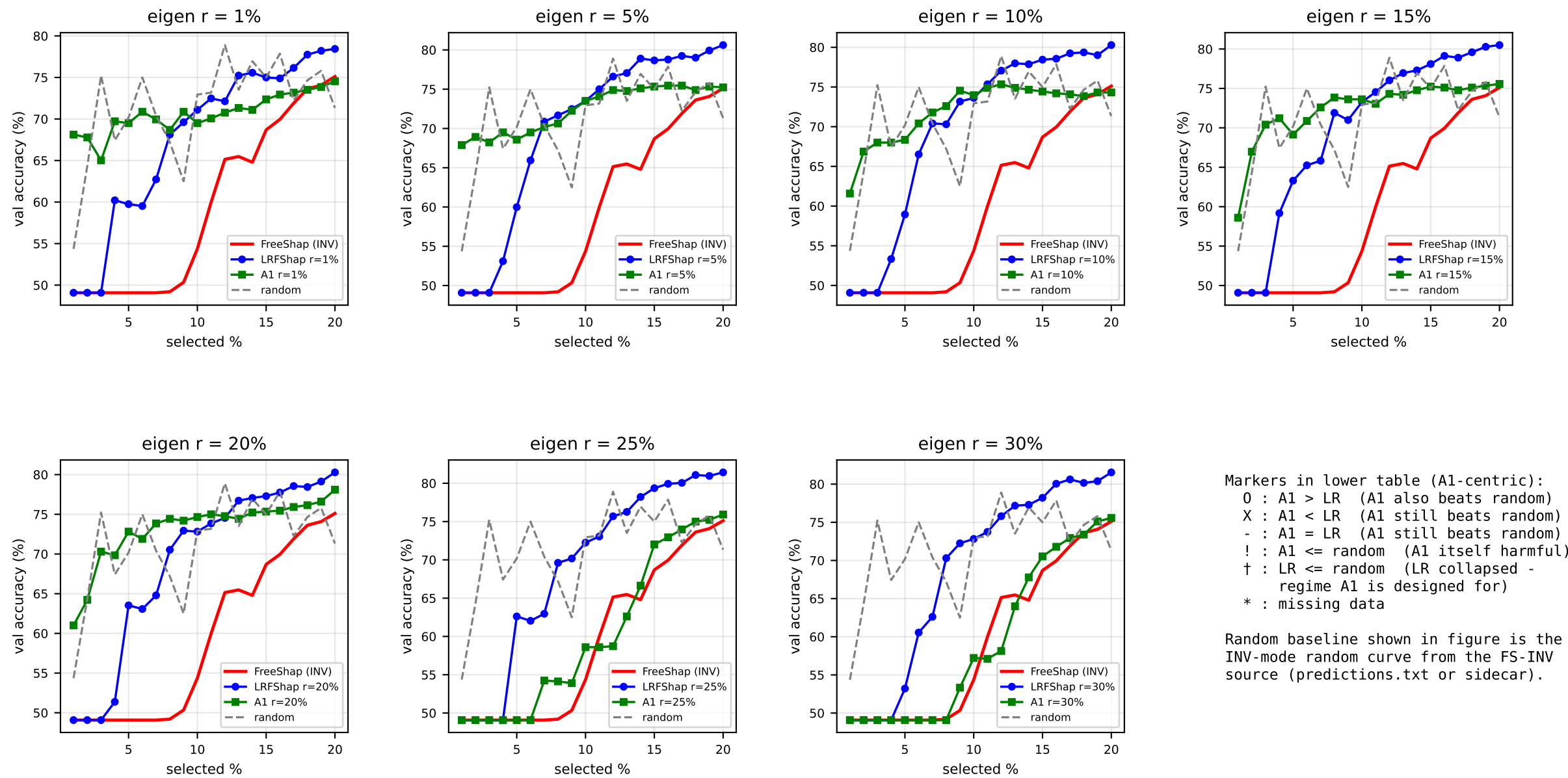
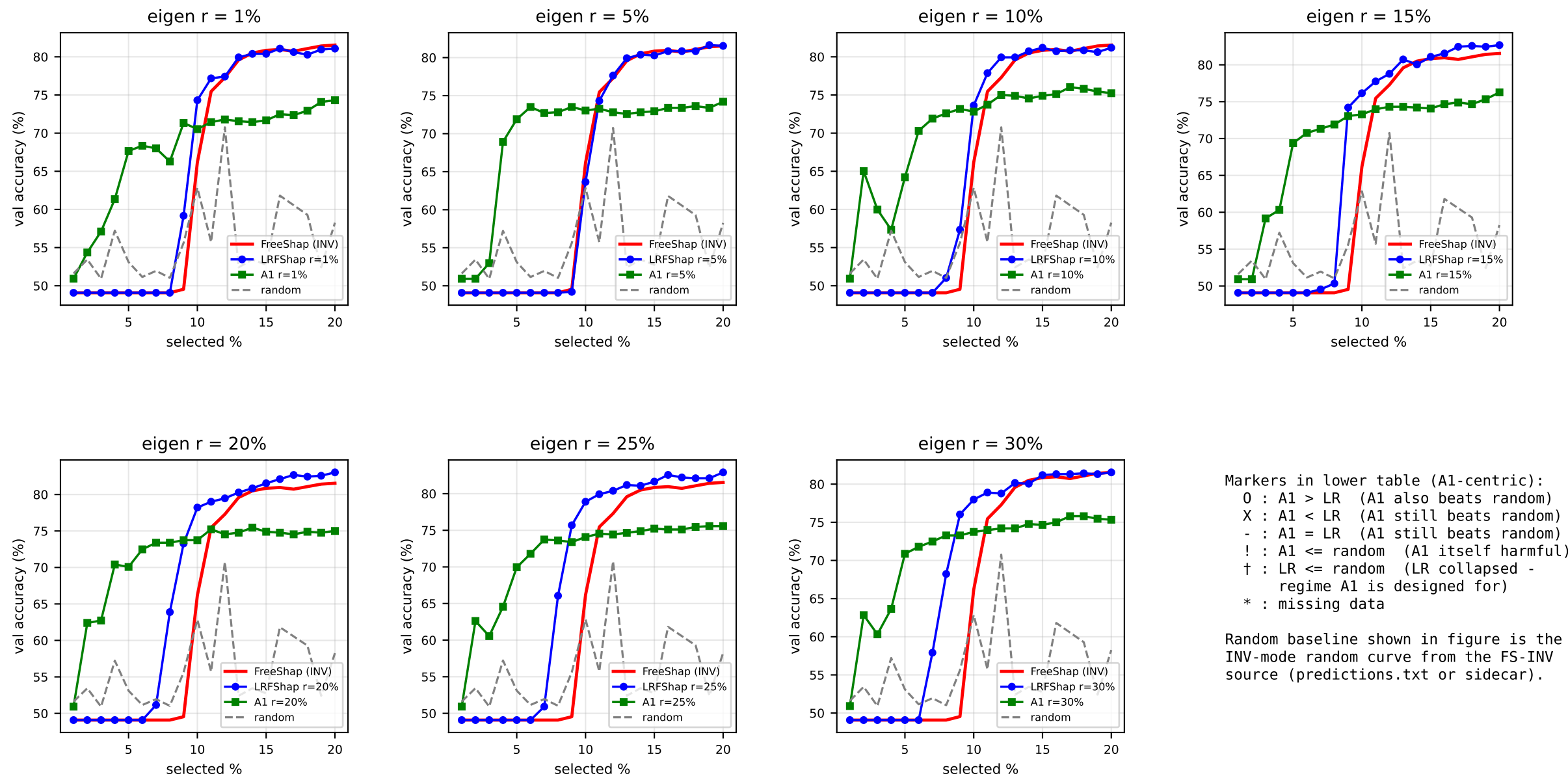




	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	49.08 [!]	49.08 [!]	54.70 [!]	70.18 [!]	74.08 [!]	79.24 [!]	80.73 [!]
r=1%	X	!	!†	O†	!†	!†	!†
r=5%	O	!	!	X	!†	!†	!
r=10%	X	X	!	X	!	!†	!
r=15%	O	O†	!	X	X	!	!
r=20%	O	O†	O	X	X	X	X
r=25%	O	O†	O	X	!	!	!
r=30%	O†	X	!	X	X	!	X



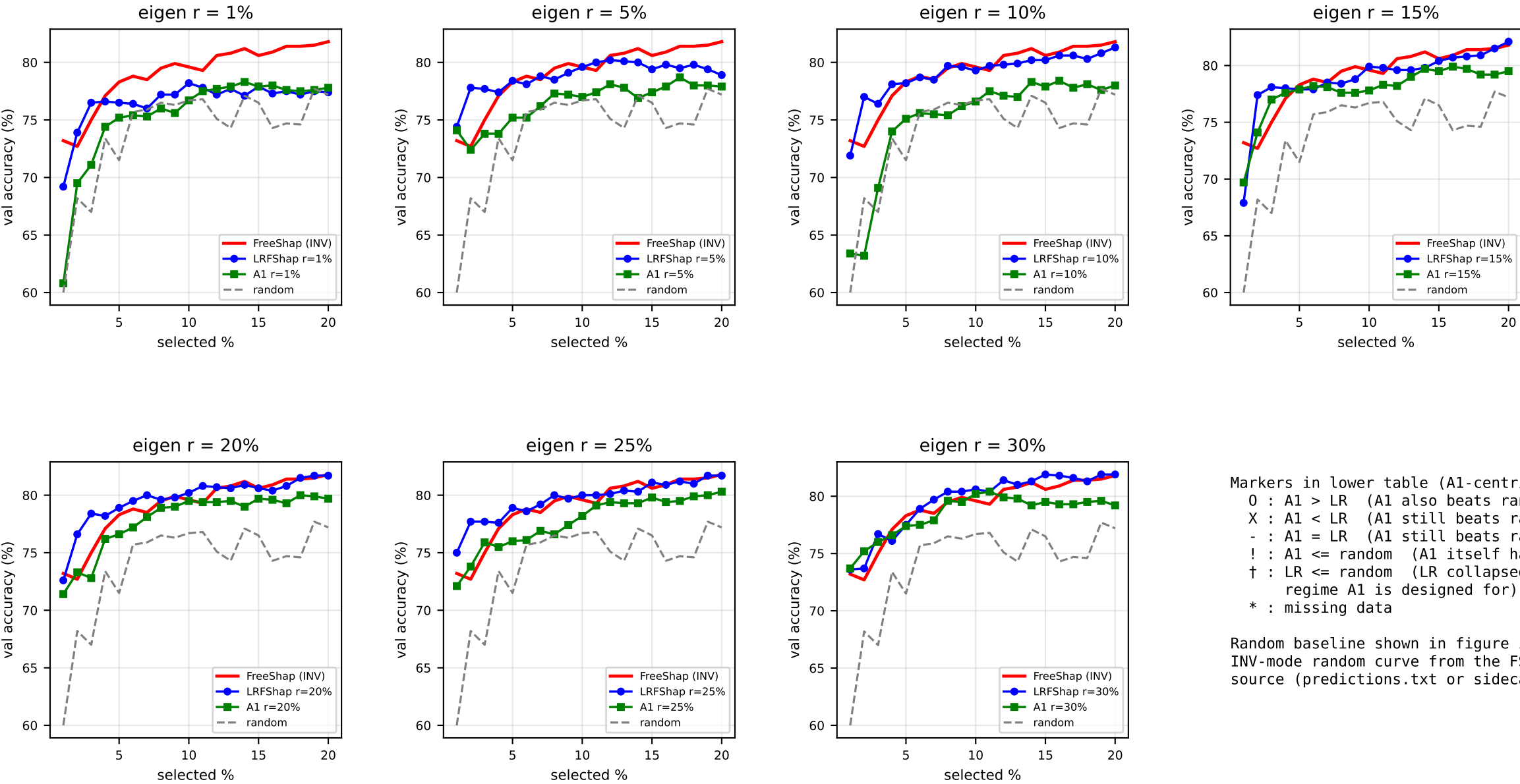
	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	49.08 [!]	49.08 [!]	49.08 [!]	49.08 [!]	49.08 [!]	54.36 [!]	75.11
r=1%	O†	!†	!†	!†	!†	!†	!
r=5%	O†	!†	!†	!†	!†	-	X
r=10%	O†	!†	!†	!†	!†	O	!
r=15%	!†	!†	!†	!†	!†	O	X
r=20%	O†	!†	!†	!†	!†	O	X
r=25%	!†	!†	!†	!†	!†	!	X
r=30%	!†	!†	!†	!†	!†	!	X



Markers in lower table (A1-centric):
0 : A1 > LR (A1 also beats random)
X : A1 < LR (A1 still beats random)
- : A1 = LR (A1 still beats random)
! : A1 <= random (A1 itself harmful)
† : LR <= random (LR collapsed - regime A1 is designed for)
* : missing data

Random baseline shown in figure is the INV-mode random curve from the FS-INV source (predictions.txt or sidecar).

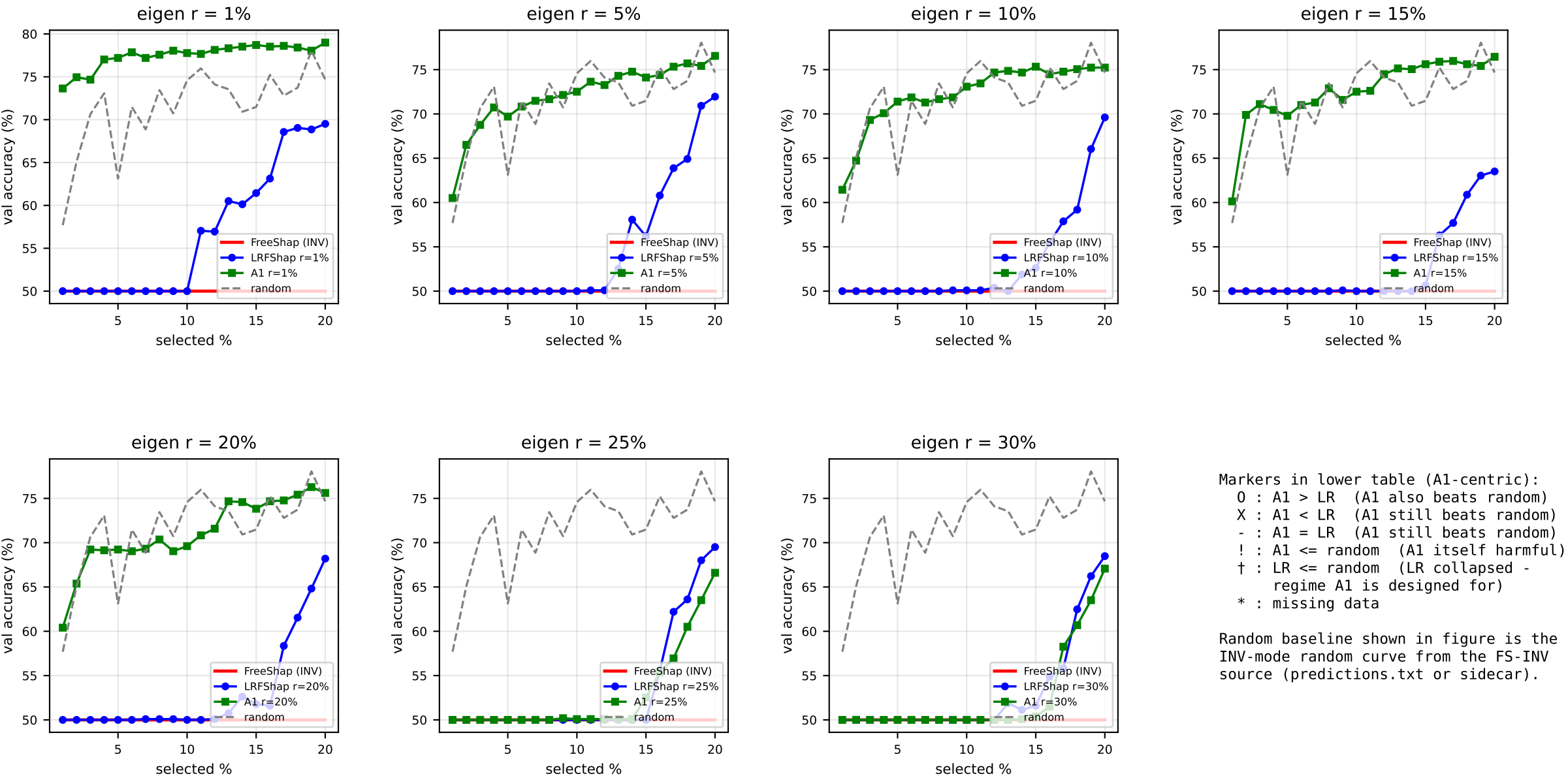
	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	49.08 [!]	49.08 [!]	49.08 [!]	49.08 [!]	49.08 [!]	66.17	81.54
r=1%	!†	O†	O†	O†	O†	X	X
r=5%	!†	!†	O†	O†	O†	O	X
r=10%	!†	O†	O†	O†	O†	X	X
r=15%	!†	!†	O†	O†	O†	X	X
r=20%	!†	O†	O†	O†	O†	X	X
r=25%	!†	O†	O†	O†	O†	X	X
r=30%	!†	O†	O†	O†	O†	X	X



Markers in lower table (A1-centric):
O : A1 > LR (A1 also beats random)
X : A1 < LR (A1 still beats random)
- : A1 = LR (A1 still beats random)
! : A1 <= random (A1 itself harmful)
† : LR <= random (LR collapsed - regime A1 is designed for)
* : missing data

Random baseline shown in figure is the INV-mode random curve from the FS-INV source (predictions.txt or sidecar).

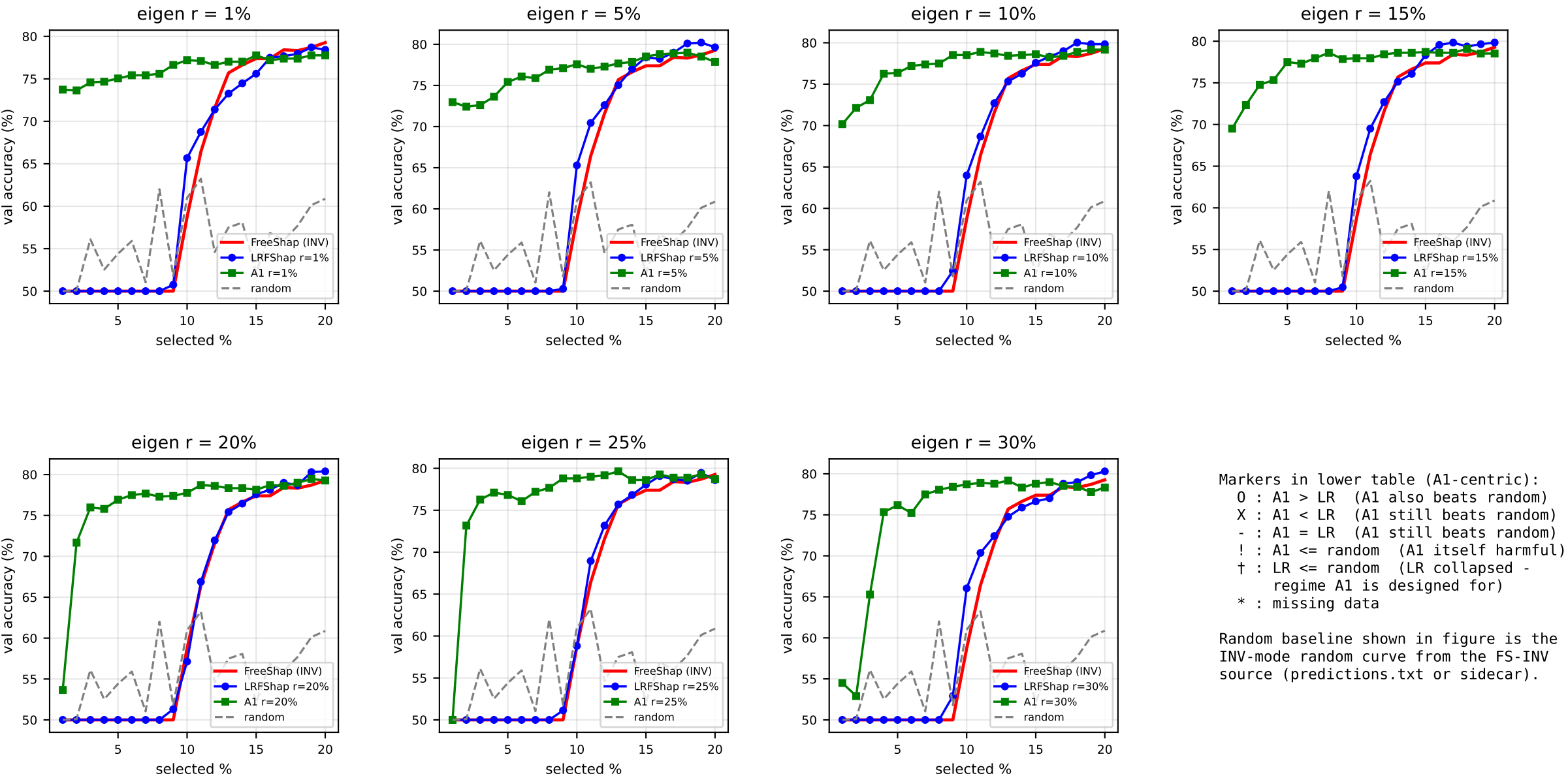
	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	73.20	72.70	75.00	77.10	78.30	79.60	81.80
r=1%	!†	X	X	X	!†	!	O†
r=5%	X	X	X	!	!	X	X
r=10%	!	!	!	!	!	!	X
r=15%	!†	X	X	X	-	X	X
r=20%	X	X	X	X	!	X	X
r=25%	X	X	X	X	!	X	X
r=30%	O	O	X	O	X	X	X



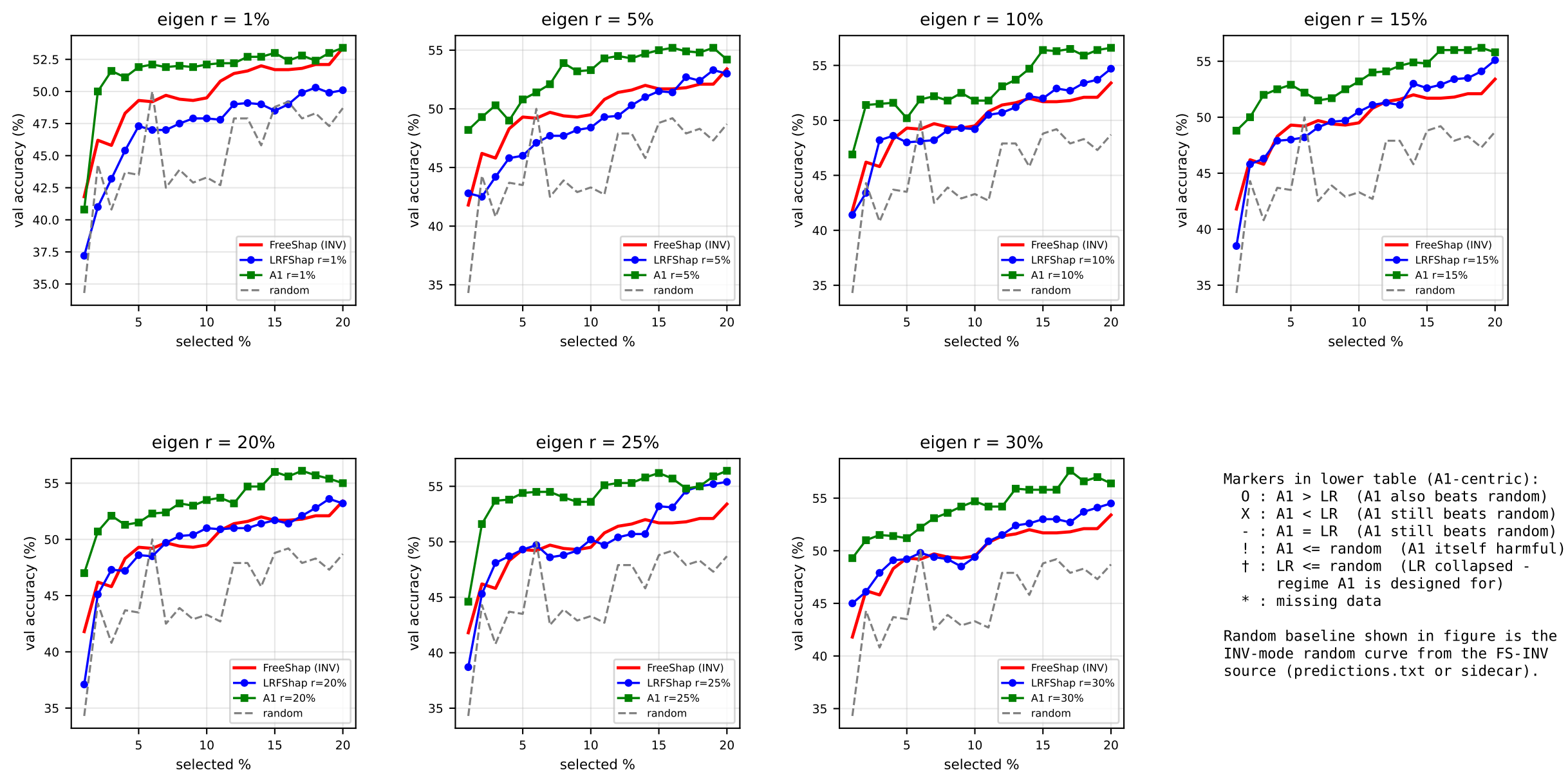
Markers in lower table (A1-centric):
O : A1 > LR (A1 also beats random)
X : A1 < LR (A1 still beats random)
- : A1 = LR (A1 still beats random)
! : A1 <= random (A1 itself harmful)
† : LR <= random (LR collapsed - regime A1 is designed for)
* : missing data

Random baseline shown in figure is the INV-mode random curve from the FS-INV source (predictions.txt or sidecar).

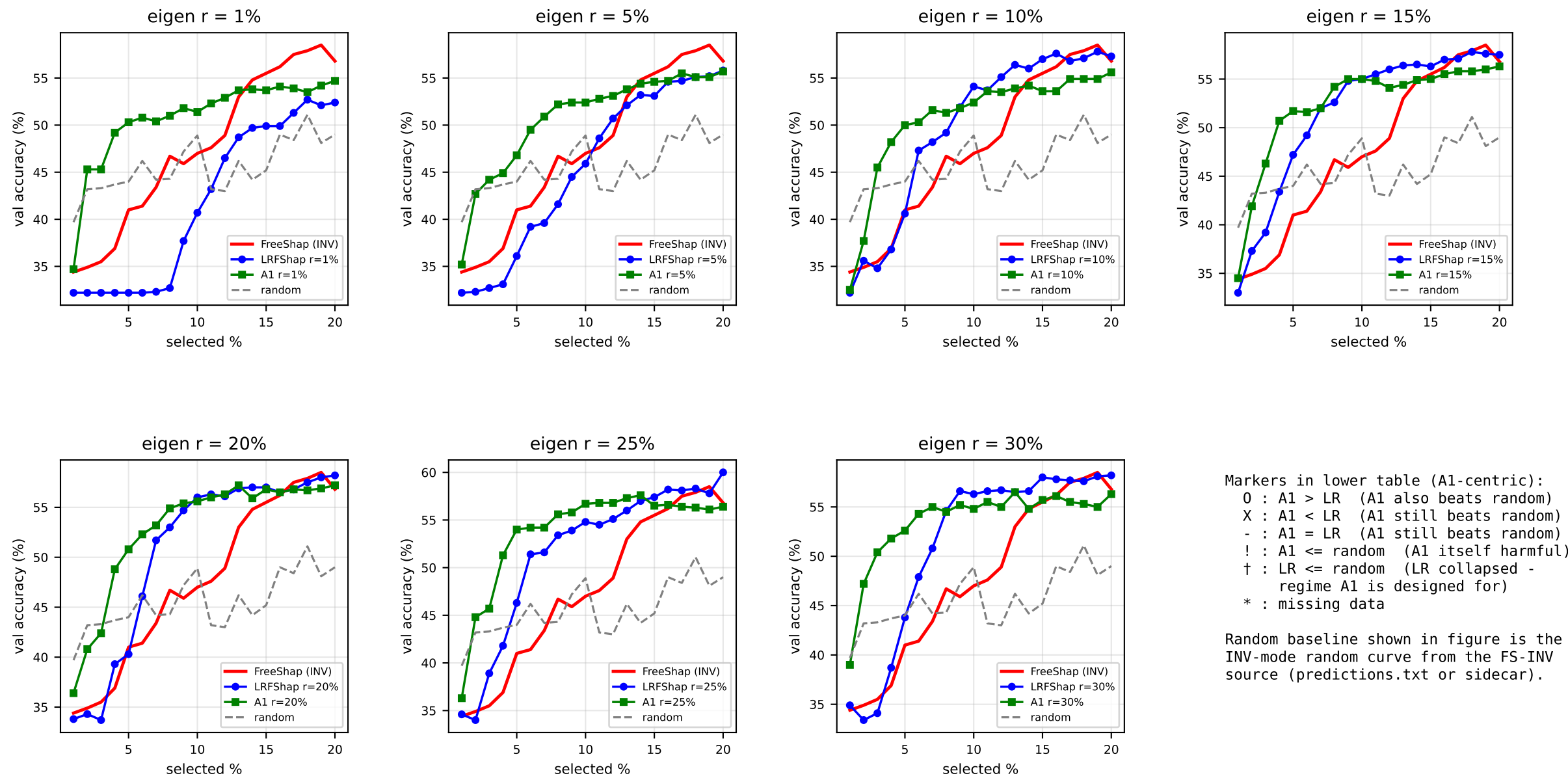
	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]
r=1%	O†	O†	O†	O†	O†	O†	O†
r=5%	!†	O†	!†	O†	!†	O†	O†
r=10%	!†	O†	!†	!†	O†	O†	O†
r=15%	!†	O†	!†	O†	!†	O†	O†
r=20%	!†	O†	!†	!†	!†	!†	O†
r=25%	!†	!†	!†	!†	!†	!†	!†
r=30%	!†	!†	!†	!†	!†	!†	!†



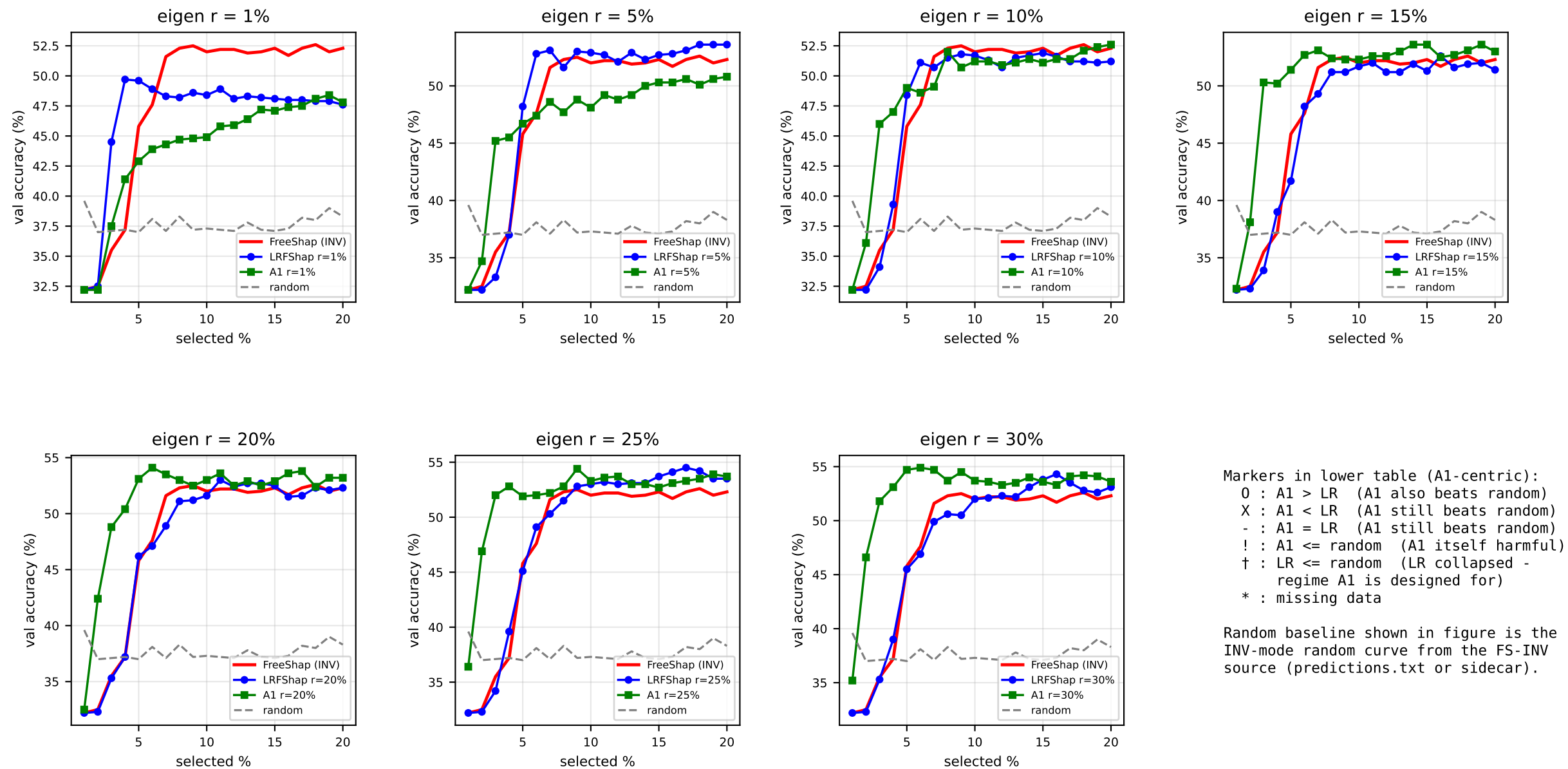
	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	58.72 [!]	79.27
r=1%	O†	O†	O†	O†	O†	O	X
r=5%	O†	O†	O†	O†	O†	O	X
r=10%	O†	O†	O†	O†	O†	O	X
r=15%	O†	O†	O†	O†	O†	O	X
r=20%	!†	O†	O†	O†	O†	O	X
r=25%	!†	O†	O†	O†	O†	O	O
r=30%	!†	O†	O†	O†	O†	O	X



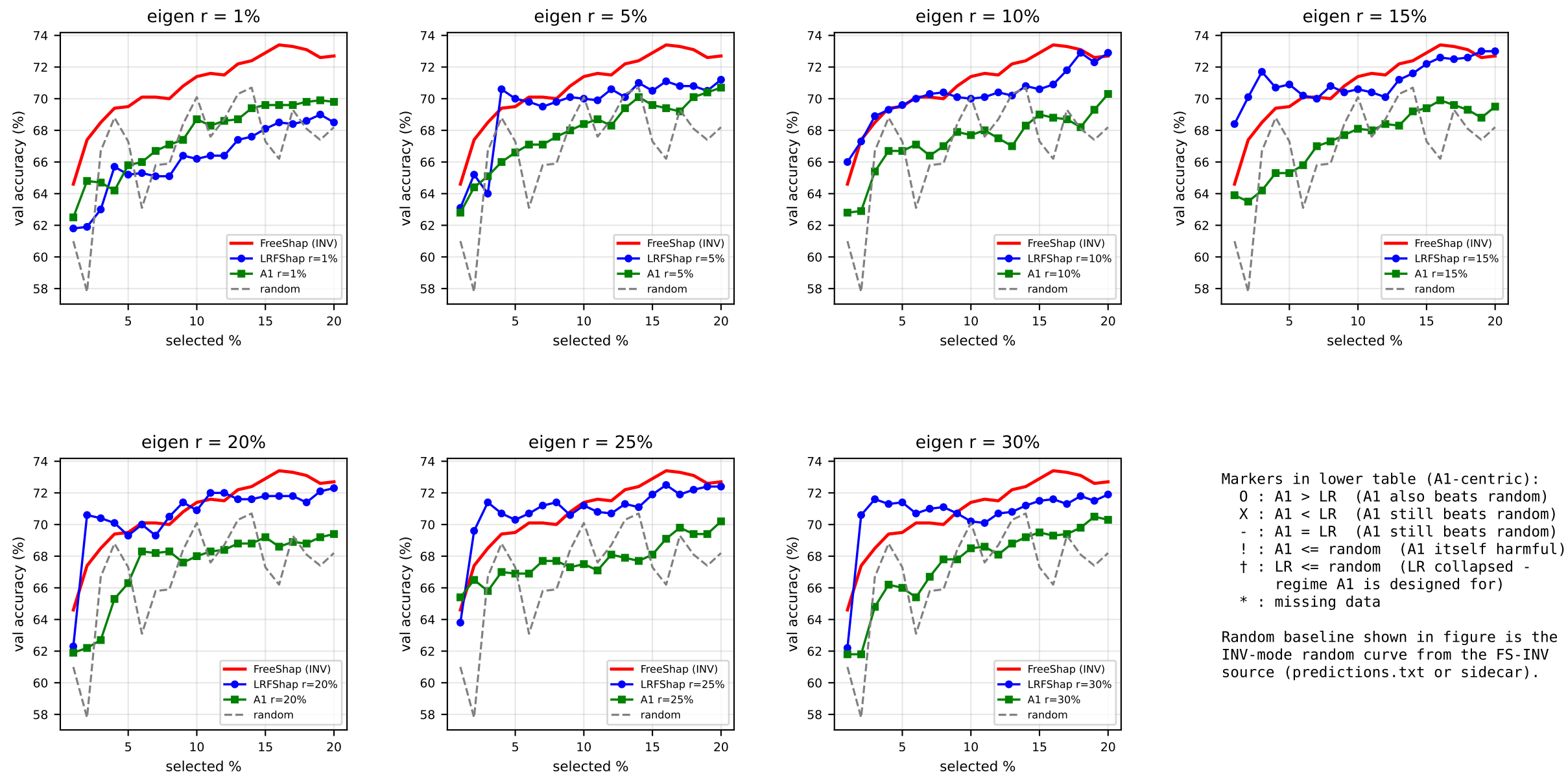
	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	41.80	46.20	45.80	48.30	49.30	49.50	53.40
r=1%	O	O	O	O	O	O	O†
r=5%	O	O	O	O	O	O	O
r=10%	O	O	O	O	O	O	O
r=15%	O	O	O	O	O	O	O
r=20%	O	O	O	O	O	O	O
r=25%	O	O	O	O	O	O	O
r=30%	O	O	O	O	O	O	O



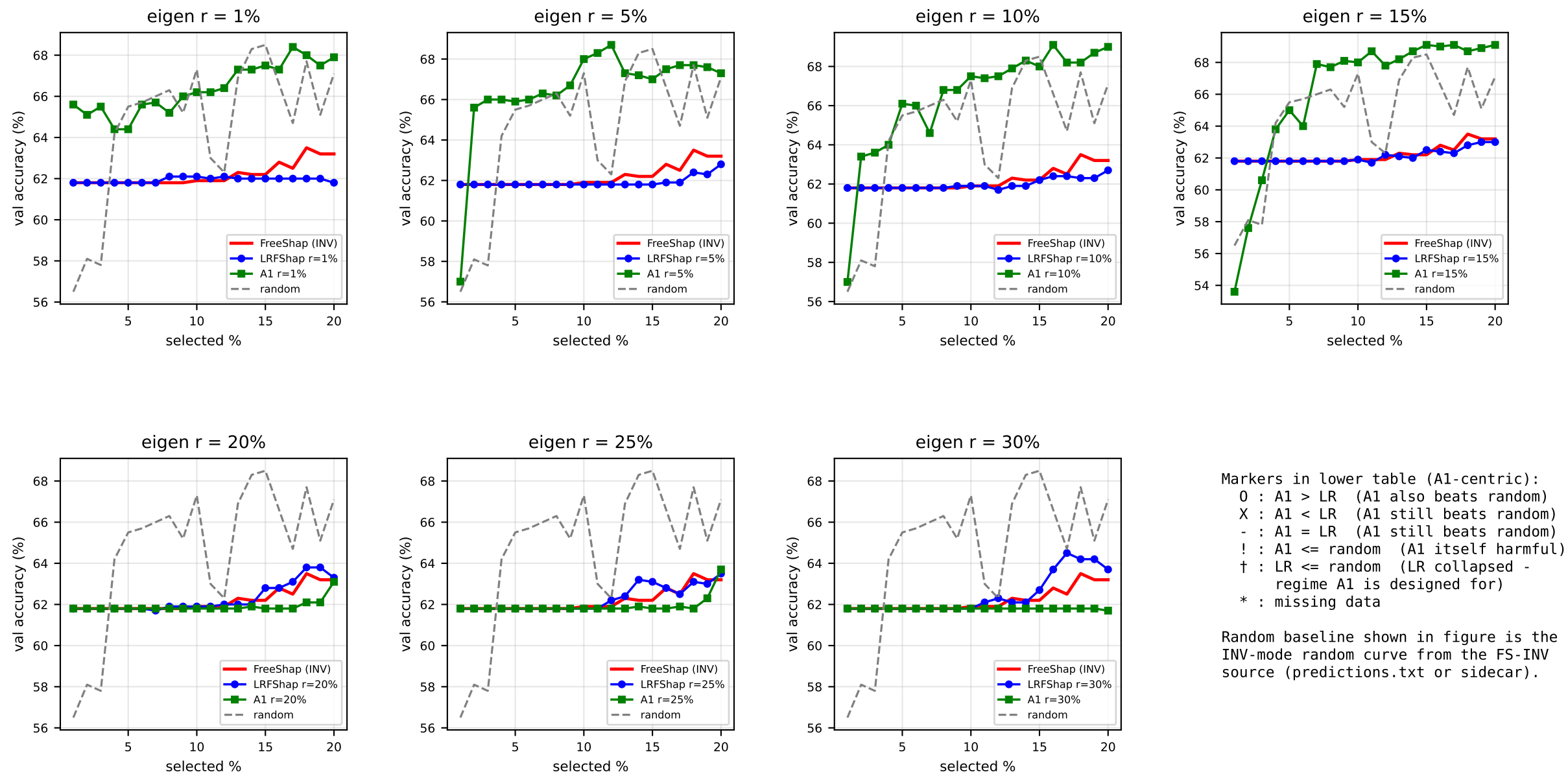
	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	34.40 [!]	34.90 [!]	35.50 [!]	36.90 [!]	41.00 [!]	47.00 [!]	56.80
r=1%	!†	O†	O†	O†	O†	O†	O
r=5%	!†	O†	O†	O†	O†	O	X
r=10%	!†	!†	O†	O†	O†	X	X
r=15%	!†	O†	O†	O	O	-	X
r=20%	!†	O†	O†	O†	O†	X	X
r=25%	!†	O†	O†	O†	O	O	X
r=30%	!†	O†	O†	O†	O†	X	X



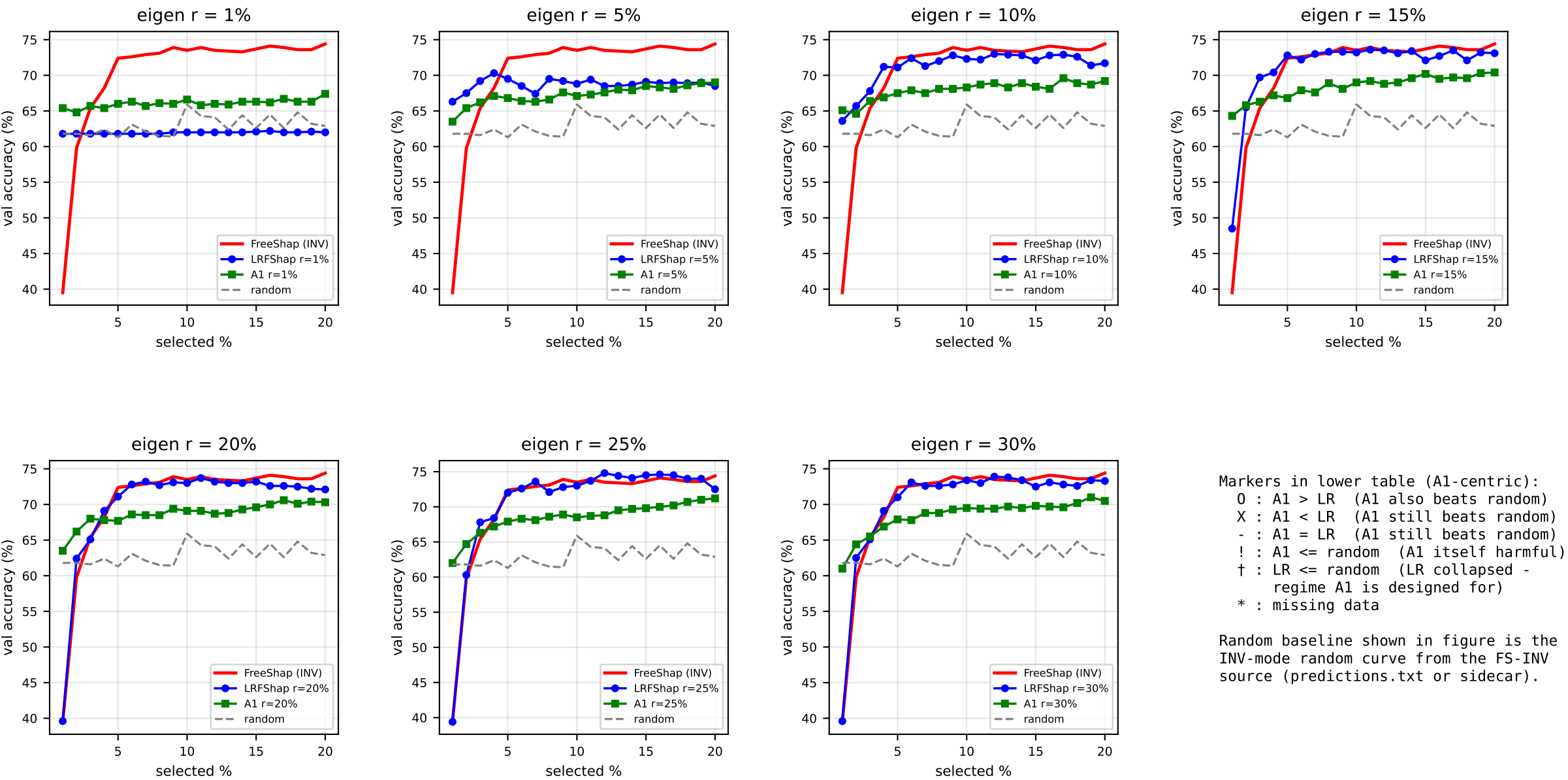
	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	32.20 [!]	32.50 [!]	35.50 [!]	37.20 [!]	45.80	52.00	52.30
r=1%	!†	!†	!	X	X	X	O
r=5%	!†	!†	O†	O†	X	X	X
r=10%	!†	!†	O†	O	O	X	O
r=15%	!†	O†	O†	O	O	O	O
r=20%	!†	O†	O†	O†	O	O	O
r=25%	!†	O†	O†	O	O	O	O
r=30%	!†	O†	O†	O	O	O	O



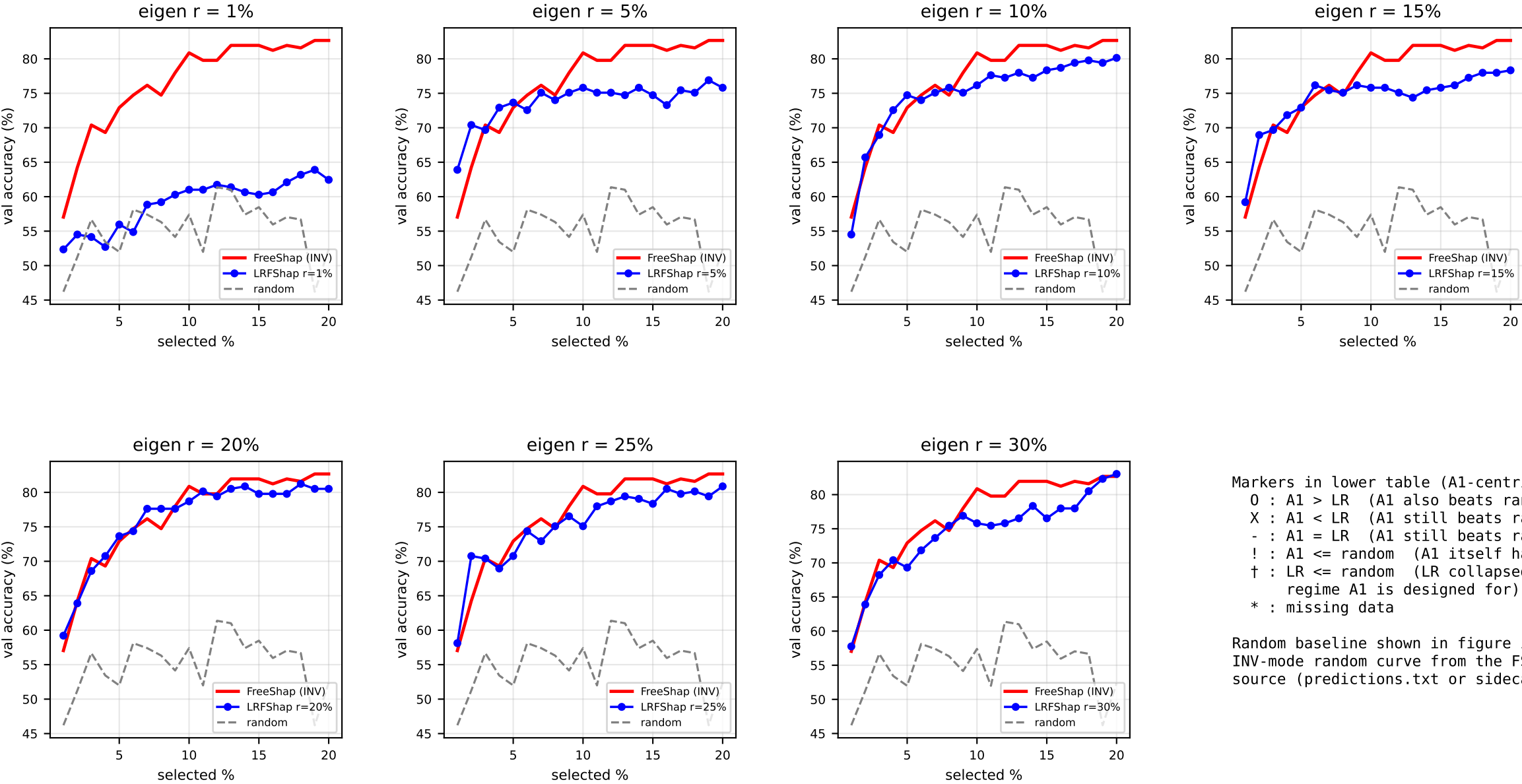
	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	64.60	67.40	68.50	69.40	69.50	71.40	72.70
r=1%	O	!†	O†	!†	O	O	O
r=5%	X	!†	O†	X	X	X	X
r=10%	X	!	X	X	X	X	X
r=15%	X	!	!	!	X	X	X
r=20%	X	!	!	!	X	X	X
r=25%	O	X	X	X	X	X	X
r=30%	X	!	X	X	X	X	X



	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	61.80	61.80	61.80	61.80 [!]	61.80 [!]	61.90 [!]	63.20 [!]
r=1%	O	O	O†	O	O†	O†	!†
r=5%	!	O	O†	O	O†	O†	!†
r=10%	!	O	!†	O	O†	O†	O†
r=15%	!	X	!†	O	O†	O†	O†
r=20%	-	-	!†	-	!†	!†	!†
r=25%	-	-	!†	-	!†	!†	!†
r=30%	-	-	!†	-	!†	!†	!†



	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	39.50 [!]	59.80 [!]	65.40	68.20	72.40	73.50	74.40
r=1%	O†	O	O†	O	O†	O†	O†
r=5%	X	X	X	X	X	X	O
r=10%	O	X	X	X	X	X	X
r=15%	O†	O	X	X	X	X	X
r=20%	O†	O	O	X	X	X	X
r=25%	!†	O†	X	X	X	X	X
r=30%	!†	O	O	X	X	X	X

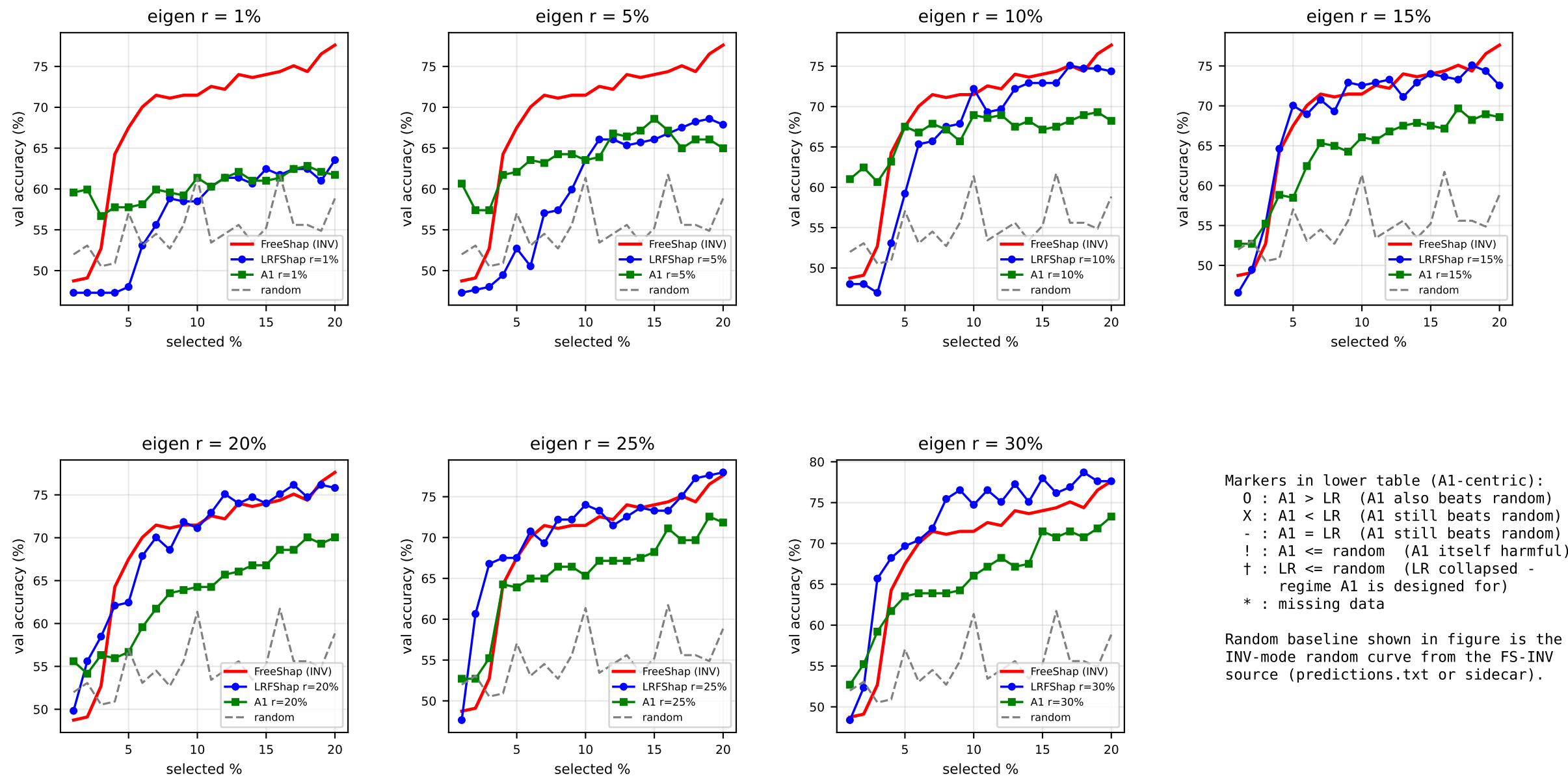


Markers in lower table (A1-centric):

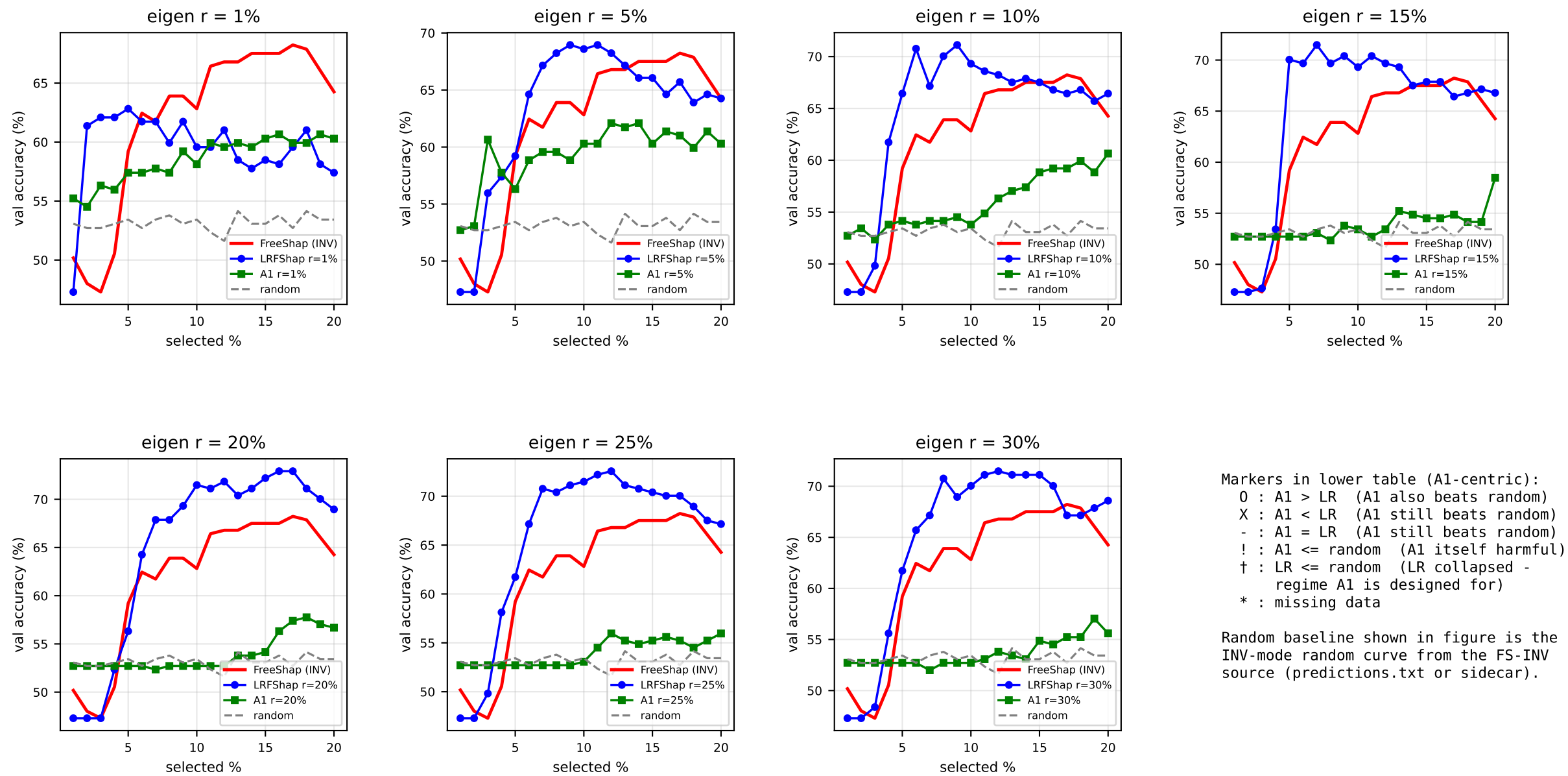
- 0 : A1 > LR (A1 also beats random)
- X : A1 < LR (A1 still beats random)
- : A1 = LR (A1 still beats random)
- ! : A1 <= random (A1 itself harmful)
- † : LR <= random (LR collapsed - regime A1 is designed for)
- * : missing data

Random baseline shown in figure is the INV-mode random curve from the FS-INV source (predictions.txt or sidecar).

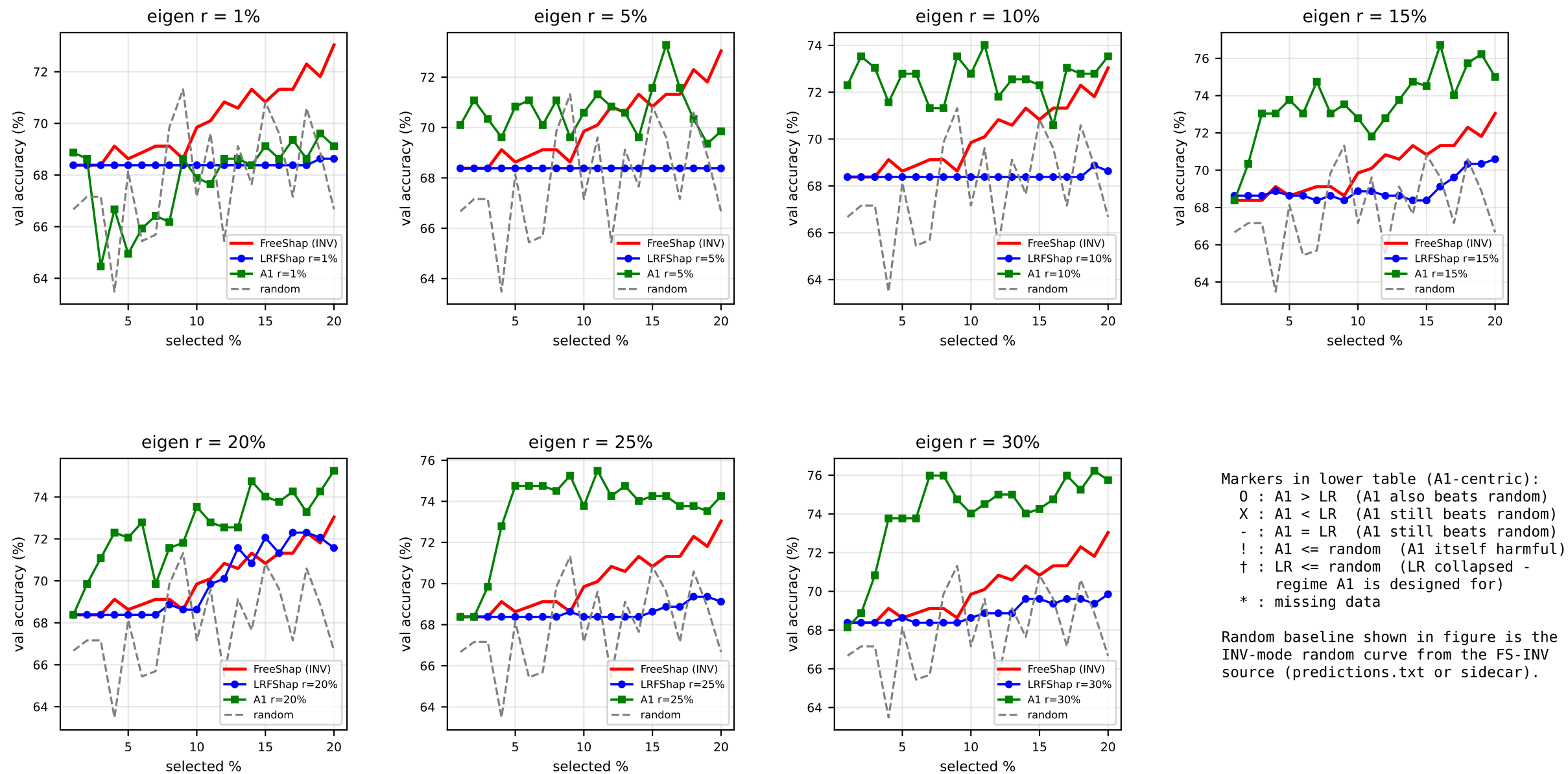
	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	57.04	64.26	70.40	69.31	72.92	80.87	82.67
r=1%	*	*	*	*	*	*	*
r=5%	*	*	*	*	*	*	*
r=10%	*	*	*	*	*	*	*
r=15%	*	*	*	*	*	*	*
r=20%	*	*	*	*	*	*	*
r=25%	*	*	*	*	*	*	*
r=30%	*	*	*	*	*	*	*



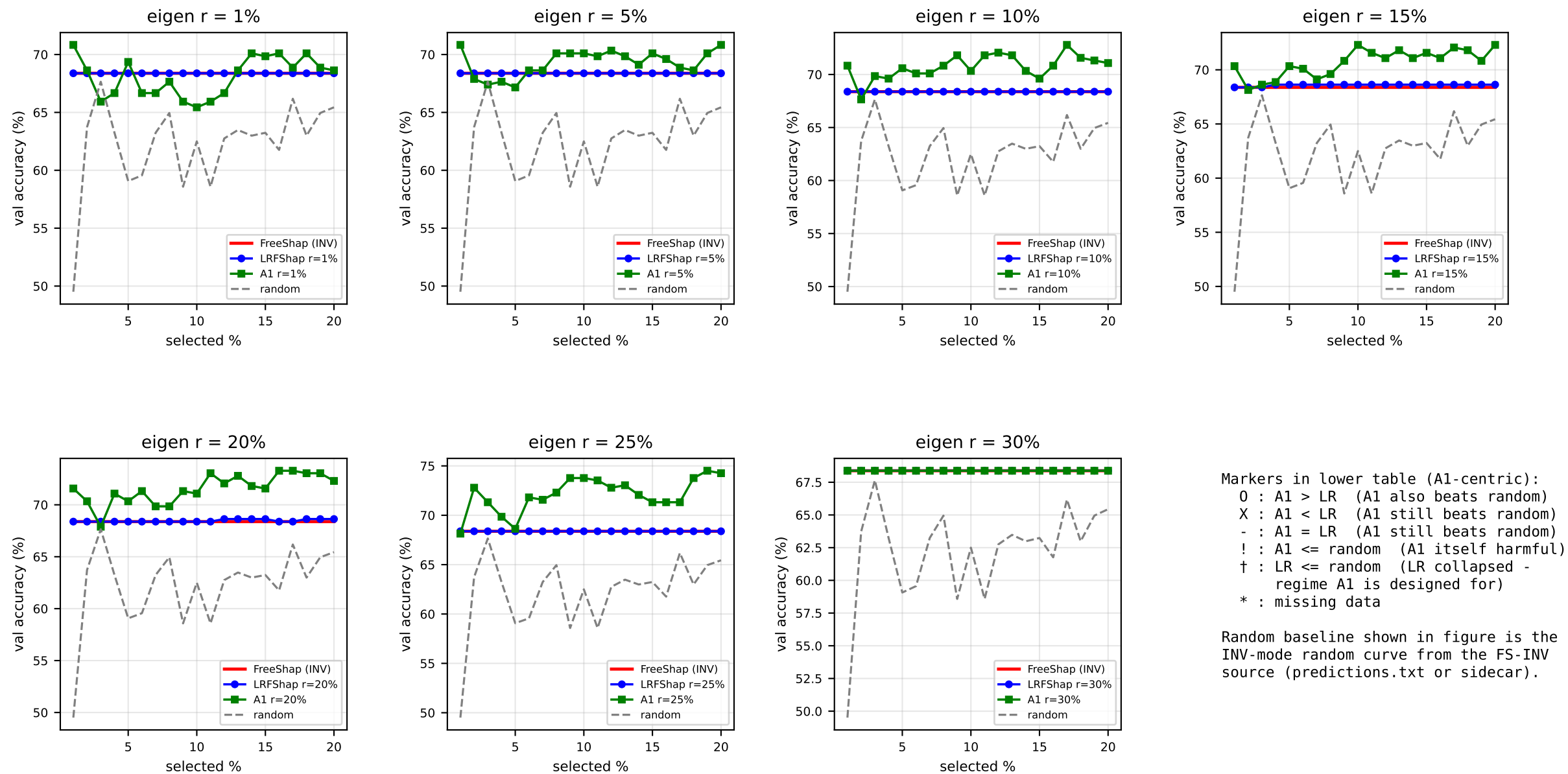
	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	48.74 [!]	49.10 [!]	52.71	64.26	67.51	71.48	77.62
r=1%	O†	O†	O†	O†	O†	O	X
r=5%	O†	O†	O†	O†	O†	-	X
r=10%	O†	O†	O†	O	O	X	X
r=15%	!†	!†	!†	X	X	X	X
r=20%	O†	!†	X	X	X	X	X
r=25%	!†	!	!	X	X	X	X
r=30%	!†	!†	X	X	X	X	X



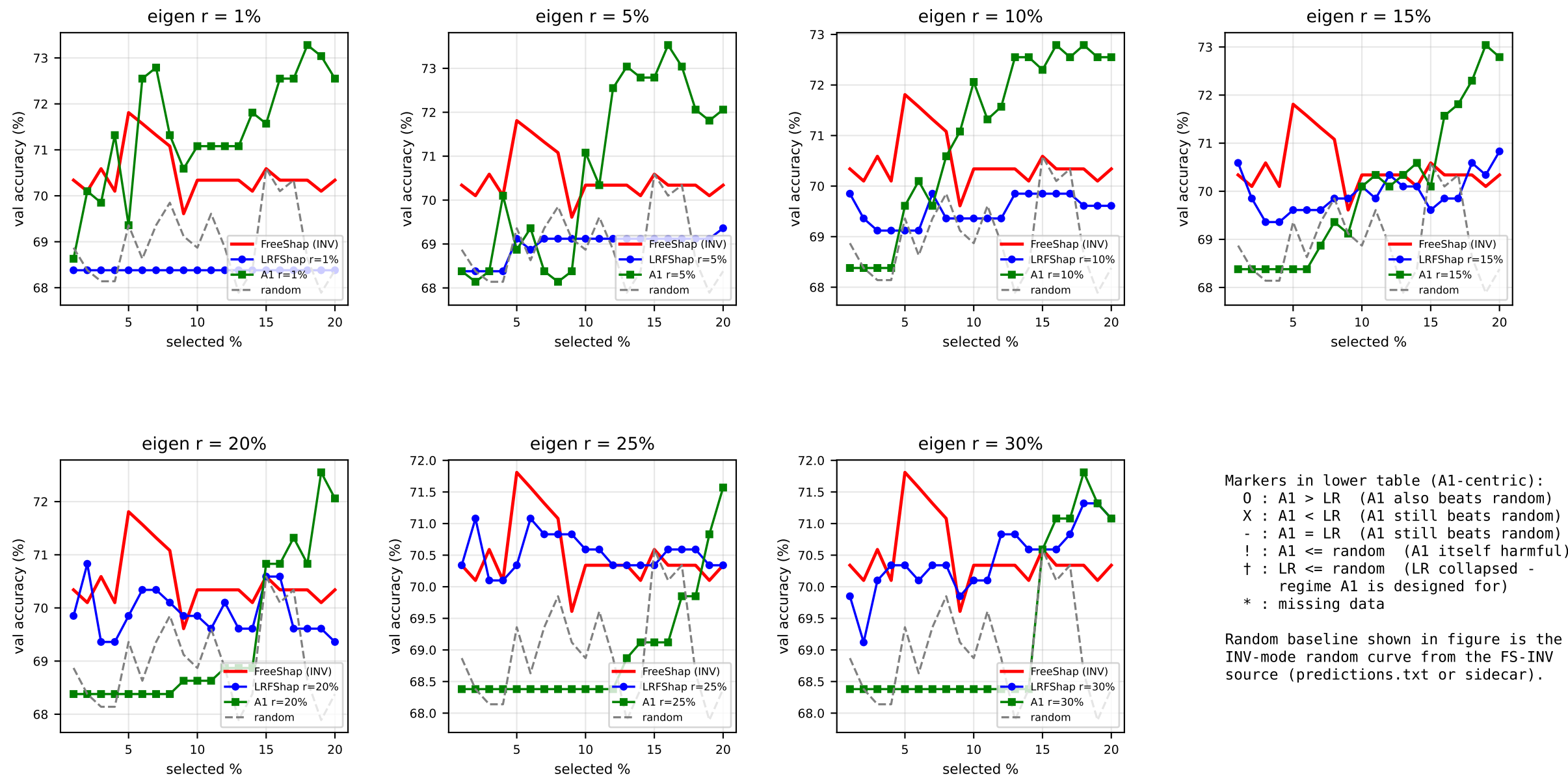
	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	50.18 [!]	48.01 [!]	47.29 [!]	50.54 [!]	59.21	62.82	64.26
r=1%	O†	X	X	X	X	X	O
r=5%	!†	O†	O	O	X	X	X
r=10%	!†	O†	O†	X	X	X	X
r=15%	!†	!†	O†	X	!	X	X
r=20%	!†	!†	O†	O	!	X	X
r=25%	!†	!†	O†	X	!	X	X
r=30%	!†	!†	O†	X	!	X	X



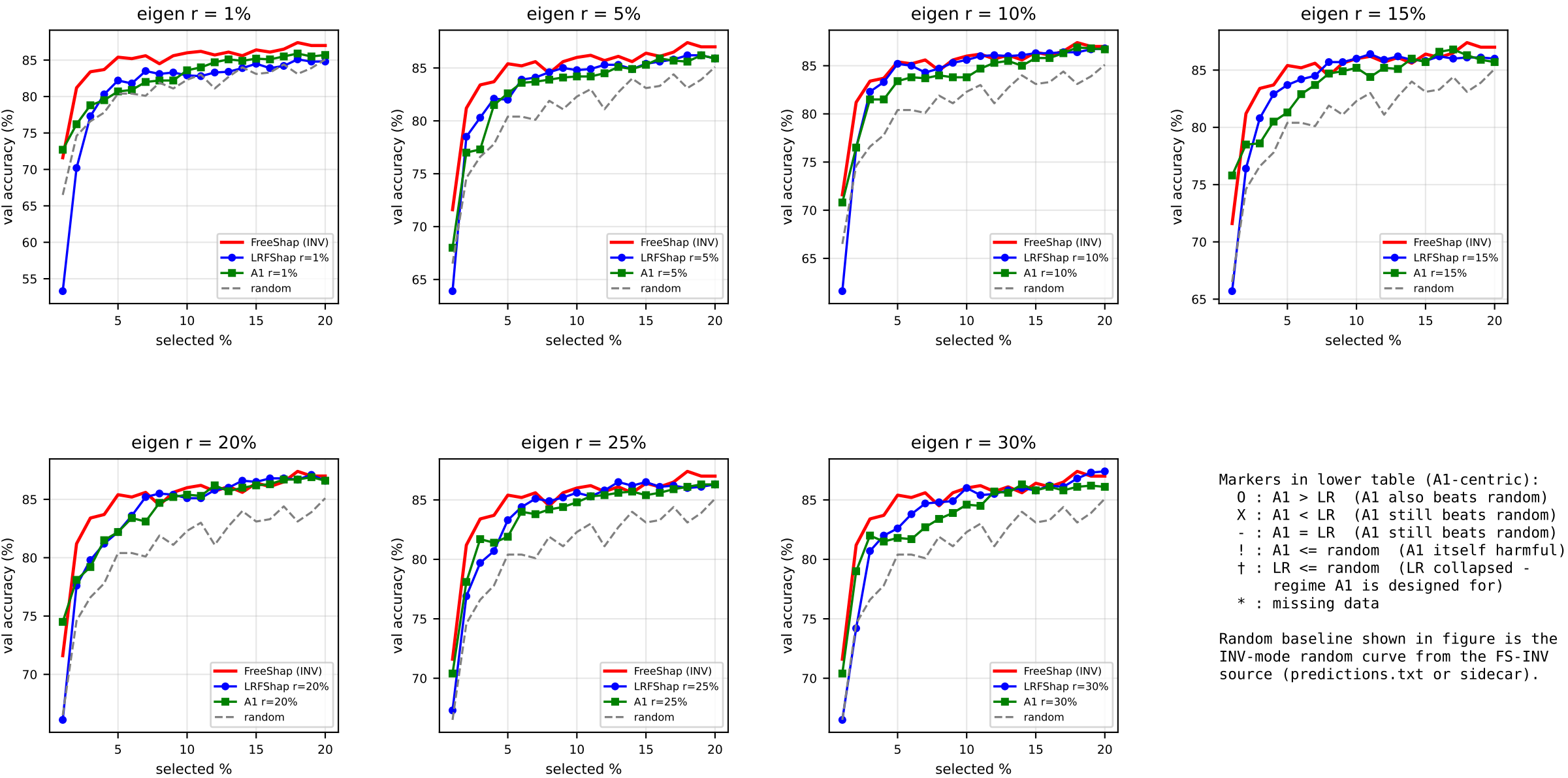
	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	68.38	68.38	68.38	69.12	68.63	69.85	73.04
r=1%	O†	O	X	!	!†	!†	!†
r=5%	O†	O	O	O	O†	!†	O†
r=10%	O†	O	O	O	O†	O†	O†
r=15%	!†	O	O	O	O†	O†	O
r=20%	!†	O	O	O	O†	O†	O
r=25%	!†	-	O	O	O†	O†	O†
r=30%	!†	O	O	O	O†	O†	O



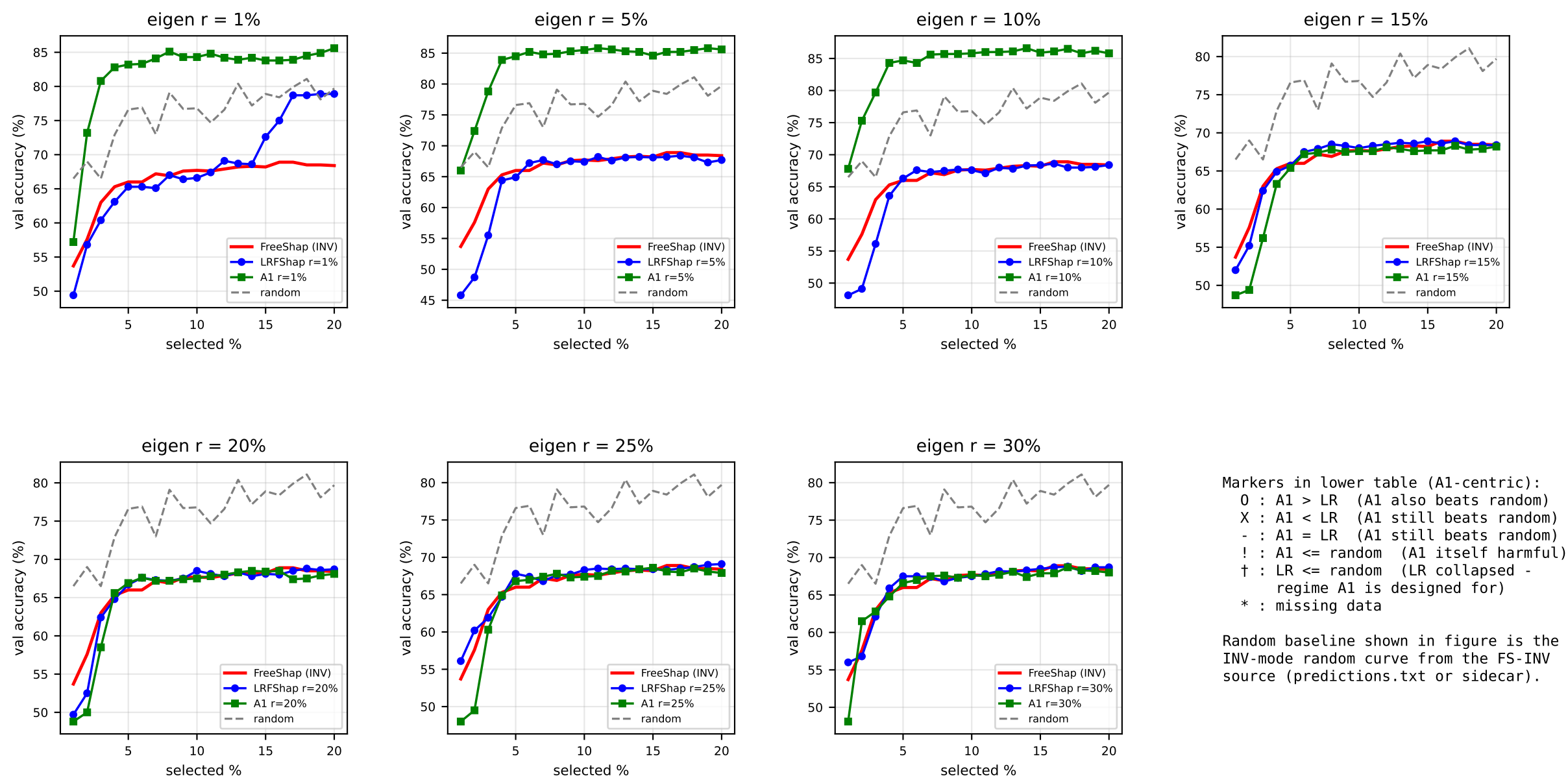
	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	68.38	68.38	68.38	68.38	68.38	68.38	68.38
r=1%	O	O	X	X	O	X	O
r=5%	O	X	X	X	X	O	O
r=10%	O	X	O	O	O	O	O
r=15%	O	X	O	O	O	O	O
r=20%	O	O	X	O	O	O	O
r=25%	X	O	O	O	O	O	O
r=30%	-	-	-	-	-	-	-



	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	70.34	70.10	70.59	70.10	71.81	70.34	70.34
r=1%	O	O	O†	O†	!†	O†	O†
r=5%	-	!	!†	O†	!†	O†	O
r=10%	X	X	!	!	O†	O†	O
r=15%	X	X	!	!	!	-	O
r=20%	X	X	!	!	!	!	O
r=25%	X	X	!	!	!	!	O
r=30%	X	X	!	!	!	!	-



	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	71.60	81.20	83.40	83.70	85.40	86.00	87.00
r=1%	O†	!†	O	!†	X	O	O
r=5%	O	X	X	X	O	X	-
r=10%	O	!†	X	X	X	X	X
r=15%	O	O†	X	!	X	X	X
r=20%	O	O	X	O†	-	O	-
r=25%	O	O	O	O†	X	X	-
r=30%	O	O†	O	X	X	X	X

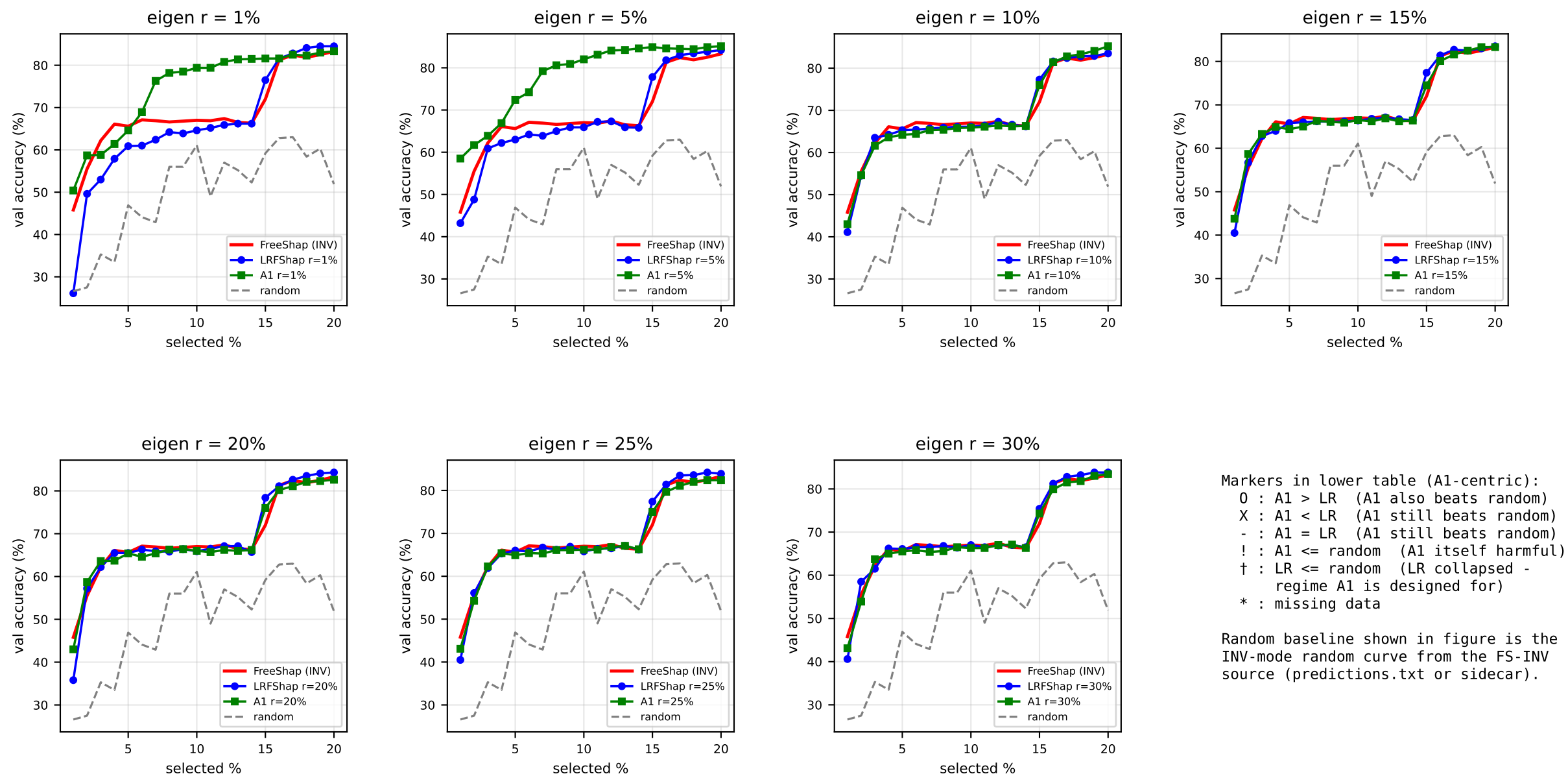


Markers in lower table (A1-centric):

- O : A1 > LR (A1 also beats random)
- X : A1 < LR (A1 still beats random)
- : A1 = LR (A1 still beats random)
- ! : A1 <= random (A1 itself harmful)
- † : LR <= random (LR collapsed - regime A1 is designed for)
- * : missing data

Random baseline shown in figure is the INV-mode random curve from the FS-INV source (predictions.txt or sidecar).

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	53.70 [!]	57.60 [!]	63.00 [!]	65.30 [!]	66.00 [!]	67.70 [!]	68.40 [!]
r=1%	O	O†	O†	O†	O†	O†	O†
r=5%	O†	O†	O†	O†	O†	O†	O†
r=10%	O†	O†	O†	O†	O†	O†	O†
r=15%	X	!†	!†	!†	!†	!†	!†
r=20%	X	!†	!†	!†	!†	!†	!†
r=25%	!	!†	!†	!†	!†	!†	!†
r=30%	!	!†	!†	!†	!†	!†	!†



Markers in lower table (A1-centric):
O : A1 > LR (A1 also beats random)
X : A1 < LR (A1 still beats random)
- : A1 = LR (A1 still beats random)
! : A1 <= random (A1 itself harmful)
† : LR <= random (LR collapsed - regime A1 is designed for)
* : missing data

Random baseline shown in figure is the INV-mode random curve from the FS-INV source (predictions.txt or sidecar).

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	45.80	55.50	62.20	66.10	65.60	67.00	83.30
r=1%	O†	O	O	O	O	O	X
r=5%	O†	O	O	O	O	O	O
r=10%	!†	O	X	X	X	X	O
r=15%	!†	O	O	O	X	O	X
r=20%	!†	O	O	X	X	O	X
r=25%	!†	X	O	O	X	O	X
r=30%	!†	X	O	X	X	X	X

Summary: A1 vs LRFShap vs random (per-cell breakdown across rank x sel%)

dataset / ratio	cells	A1 > LR	A1 < LR	A1 = LR	LR <= rnd (harmful)	A1 <= rnd (harmful)	A1 win %	LR-safe %	A1-safe %
sst2/baseline	49	11	16	0	12	22	22.4	75.5	55.1
sst2/pos70	49	7	5	1	36	36	14.3	26.5	26.5
sst2/pos90	49	27	13	0	35	9	55.1	28.6	81.6
mr/baseline	49	4	30	1	4	14	8.2	91.8	71.4
mr/pos70	49	21	0	0	49	28	42.9	0.0	42.9
mr/pos90	49	40	6	0	35	3	81.6	28.6	93.9
mnli/baseline	49	49	0	0	1	0	100.0	98.0	100.0
mnli/cls60_20_20	49	31	9	1	33	8	63.3	32.7	83.7
mnli/cls90_05_05	49	31	7	0	22	11	63.3	55.1	77.6
qqp/baseline	49	7	31	0	5	11	14.3	89.8	77.6
qqp/pos50	49	20	1	9	28	19	40.8	42.9	61.2
qqp/pos10	49	17	30	0	10	2	34.7	79.6	95.9
rte/baseline	0	0	0	0	0	0	-	-	-
rte/pos30	49	17	22	1	21	9	34.7	57.1	81.6
rte/pos10	49	12	23	0	18	14	24.5	63.3	71.4
mrpc/baseline	49	38	1	1	25	9	77.6	49.0	81.6
mrpc/pos50	49	31	11	7	0	0	63.3	100.0	100.0
mrpc/pos90	49	15	10	3	11	21	30.6	77.6	57.1
ag_news/baseline	49	20	21	4	8	4	40.8	83.7	91.8
ag_news/cls55_15_15_15	49	21	2	0	44	26	42.9	10.2	46.9
ag_news/cls85_05_05_05	49	26	18	0	7	5	53.1	85.7	89.8
ALL (21 settings)	980	445	256	28	404	251	45.4	58.8	74.4

Notes:

- n_cells = 7 ranks x 7 sel% = 49 per setting (when all data present).
- 'A1 win %' = fraction of cells where A1 strictly beats LR (within the non-harm 0/X/= subset).
- 'LR-safe %' = fraction of cells where LR > random (NOT harmful for LR).
- 'A1-safe %' = fraction of cells where A1 > random (NOT harmful for A1).
- Cells marked '!' (A1 harmful) are counted toward n_cells but contribute to a1_harm only; the 0/X/= counts use the A1-centric scheme.
- Per-cell random baseline = LR-eigen sidecar's random_results_inv at the same rank, with fallback to FS-INV random when missing.